Reader: But it is self-evident.

Author: I agree. However, this "self-evidence" is connected to one more very fine aspect without which the very calculus could not be developed. As you probably noted, one cannot identify on the real line two neighbouring points. If one point is chosen, it is impossible, in principle, to point out its "neighbouring" point. In other words, no matter how carefully you select a pair of points on the real line, it is always possible to find any number of points between the two. ↳ V. Imp

• Take, for example, the interval $[0,1]$. Now, exclude the point 1. You will have a half-open interval $[0,1)$. Can you identify the largest number over this interval?

Reader: No, it is impossible.

Author: That's right. However, if there were a point neighbouring 1, after the removal of the latter this "neighbour" would have become the largest number. I would like to note here that many "delicate" points and many "secrets" in the calculus theorems are ultimately associated with the impossibility of identifying two neighbouring points on the real line, or of specifying the greatest or least number on an open interval of the real line.

But let us get back to the properties of convergent sequences and prove the following:

Theorem: If sequences (y_n) and (z_n) are convergent (we denote their limits by a and b, respectively), a sequence (y_n+z_n) is convergent too, its limit being a+b.

Reader: This theorem is none other than rule (3) discussed in the previous dialogue.

Author: That's right. Nevertheless, I suggest you try to prove it.

Sam: **Proof:** Given, the sequence y_n is convergent to a limit a.

By definition (1) in dialogue 2, $\forall \epsilon > 0, \exists N \in$ Nat no, s.t. $\forall n \in$ Nat no, $n > N$, $|y_n - a| < \epsilon$. Take any arbitrary $\epsilon > 0$. $\therefore \exists N_1 \in N$ s.t. $\forall n > N_1, n \in N$,

$|y_n - a| < \epsilon$ (Eqn 1) $\Rightarrow$ $a-\epsilon < y_n < a+\epsilon$ (Eqn 2)

Given the sequence z_n is convergent to a limit b. Take the same ϵ.

$\therefore \exists N_2 \in N$ s.t. $\forall n > N_2, n \in N$, $|z_n - b| < \epsilon$. (Eqn 3)